Font & String

ThS. Trần Anh Dũng

fppt.com

Nội dung chính

1. Unicode
2. Font
3. String

Bộ ký tự Unicode

- Unicode là 1 bộ ký tự 16 bit chứa tất cả các ký tự thường dùng trong xử lý thông tin.
- Unicode là 1 sự đồng ý về cách lưu trữ ký tự, 1 chuẩn được sự hỗ trợ bởi các thành viên của cộng đồng Unicode
- Unicode cung cấp 1 số duy nhất cho mỗi ký tự, không liên quan đến hệ điều hành, chương trình hay ngôn ngữ nào.

Ví dụ bộ ký tự Unicode

0100 **Latin Extended-A** **017F**

	010	011	012	013	014	015	016	017
0	Ā 0100	Đ 0110	Ġ 0120	İ 0130	l· 0140	Ő 0150	Š 0160	Ű 0170
1	ā 0101	đ 0111	ġ 0121	ı 0131	Ł 0141	ő 0151	š 0161	ű 0171
2	Ă 0102	Ē 0112	Ģ 0122	IJ 0132	ł 0142	Œ 0152	Ţ 0162	Ų 0172
3	ă 0103	ē 0113	ġ 0123	ij 0133	Ń 0143	œ 0153	ţ 0163	ų 0173
4	Ą 0104	Ĕ 0114	Ĥ 0124	Ĵ 0134	ń 0144	Ŕ 0154	Ť 0164	Ŵ 0174

Định nghĩa Font

- Tập hợp hoàn chỉnh:
  - Các chữ cái
  - Các dấu câu
  - Các con số
  - Các ký tự đặc biệt
- Theo một - kiểu loại
  - Trọng lượng (thường hoặc đậm nét)
  - Dáng bộ (thẳng hoặc nghiêng)
- Với kích cỡ phù hợp và có thể phân biệt khác nhau.

Định nghĩa Font

- Một font có thể được cung cấp bởi một hoặc nhiều trang mã
- Mỗi font thường được lưu trong một file font
- Mỗi font thường được đặt một tên.
- Ví dụ font Times New Roman Bold, font Courier New Italic

Kiểu Font và Họ Font

Tên Font = Họ Font + (Kiểu Font)

Font name = Font family + (Typestyle)

Ví dụ: Họ font Times New Roman chứa 4 font khác nhau:

- Times New Roman
- Times New Roman Italic
- Times New Roman Bold
- Times New Roman Bold Italic

Kiểu Font và Họ Font

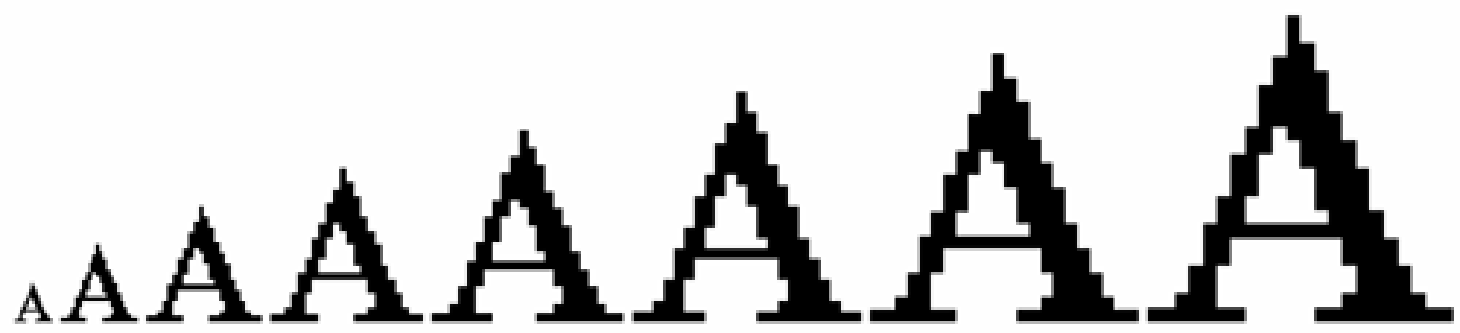
Ví dụ:

Roman	Roman	Roman	Roman
Swiss	Swiss	Swiss	Swiss
Modern	Modern	Modern	Modern
Script	Script	Script	Script
Decorative	Decorative	Decorative	DECORATIVE

Các loại Font trên HĐH Windows

- Bitmap Font
- Vector Font
- TrueType Font
- OpenType Font

Bitmap Font



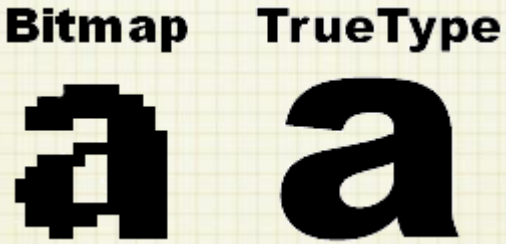
- Dùng hình ảnh để hiện thị các điểm ảnh của 1 ký tự
- Thường là những file có dạng *.fon
- Khi hiển thị ký tự có kích thước lớn thường bị nát hình

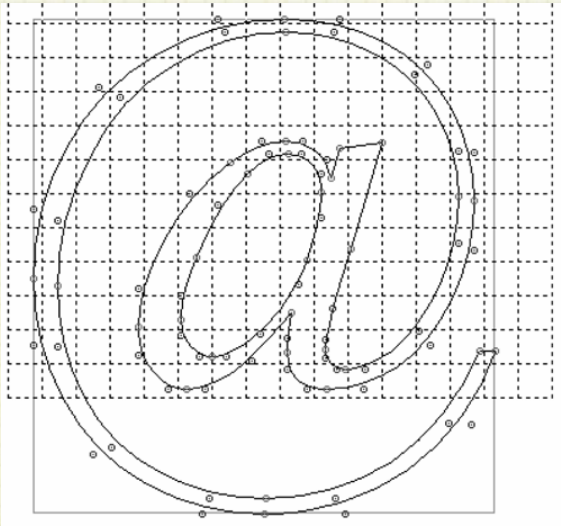
Vector Font

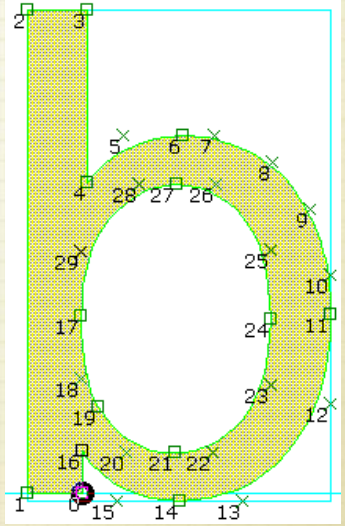
- Dùng các đoạn thẳng nối với nhau để hiển thị ký tự
- Thường là những file có dạng *.fnt
- Mặc dù hiển thị ký tự có kích thước lớn tốt hơn bitmap font nhưng vẫn chưa có được độ sắc nét cao

TrueType Font

- Dùng các đoạn thẳng và đoạn cong nối với nhau để hiển thị ký tự
- Thường là những file có dạng *.ttf
- Hiển thị ký tự có kích thước lớn với độ sắc nét cao

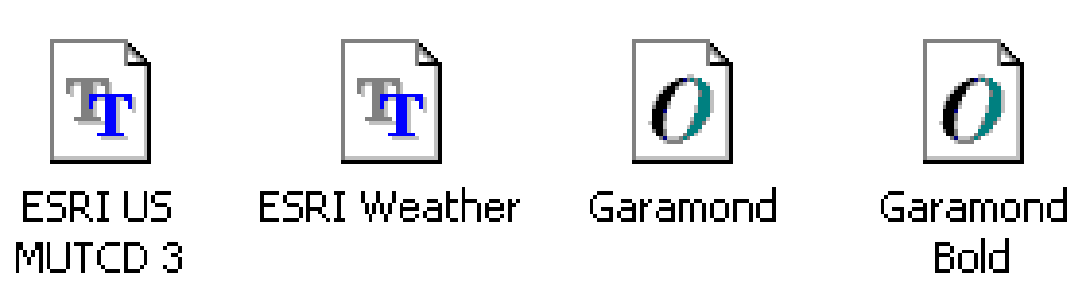




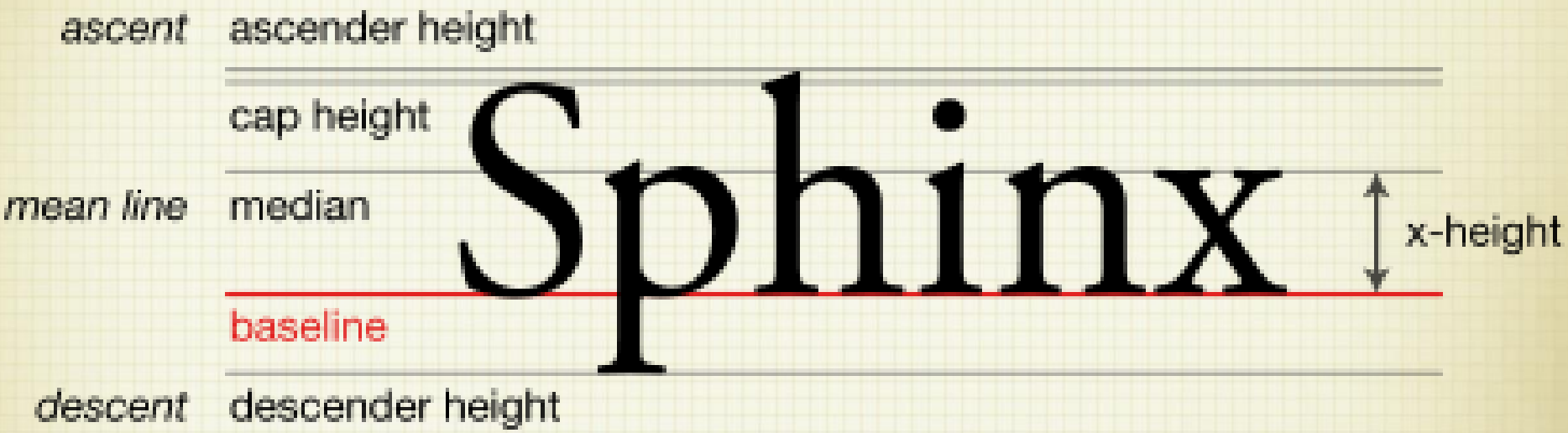


OpenType Font

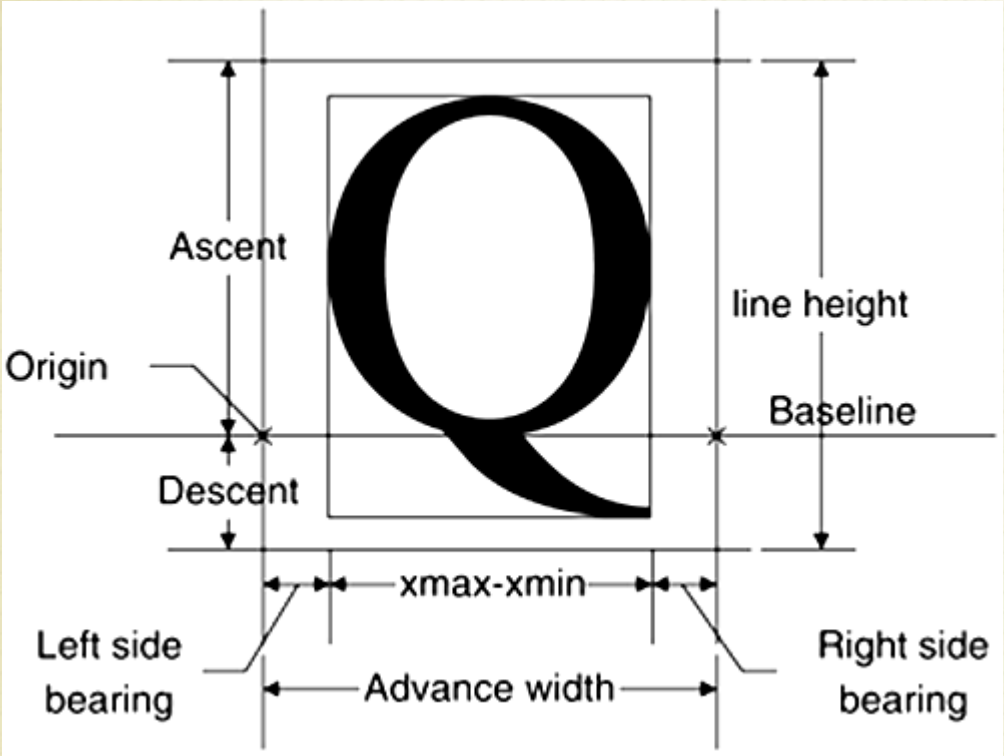
- Là chuẩn định dạng font kết hợp 2 loại định dạng font có sẵn: Type 1 (PostScript) font và TrueType font
- Độc lập với hệ điều hành (cross-platform)
- Hỗ trợ nhiều loại ngôn ngữ trong 1 font
- Thường là những file có dạng *.otf

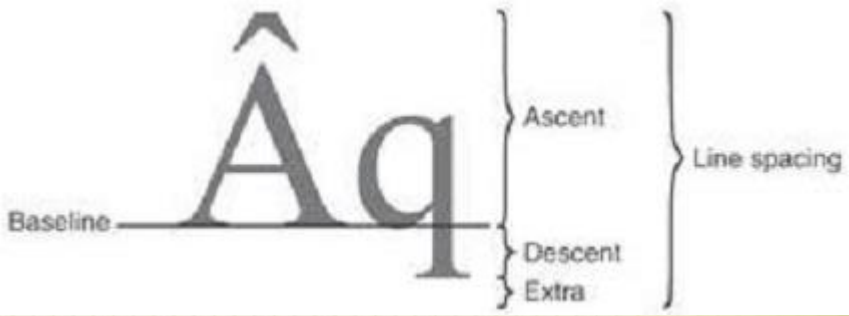


Các thông số Font (Font metrics)



Các thông số Font (Font metrics)





Độ đo Font

- Font được đo bởi nhiều độ đo:
  - pixel: phần tử nhỏ nhất của ảnh mà 1 thiết bị có thể hiển thị (màn hình, máy in)
  - point: 1 point = ? inch
  - em: độ rộng của chữ M ứng với kiểu chữ đang dùng
  - design unit: dùng để đo kích cỡ 1 họ font bằng độ đo point khi bị thay đổi kích thước

Class Font

- Mô tả lớp Font
- Các thuộc tính lớp Font
- Các hàm khởi tạo lớp Font
  - Tập hợp FontStyle
  - Tập hợp GraphicsUnit
- Các phương thức lớp Font

Mô tả lớp Font

- Dùng để xác định cách định dạng văn bản
- Bao gồm các thuộc tính cách thể hiện, kích thước, kiểu dáng
- Không cho phép kế thừa
- **Namespace:** System.Drawing
- **Assembly:** System.Drawing (in dll)

Font Properties

Name	Description
Bold	Gets a value that indicates whether this **Font** object is bold.
FontFamily	Gets the FontFamily object associated with this **Font** object.
GdiCharSet	Gets a byte value that specifies the GDI character set that this **Font** object uses.
GdiVerticalFont	Gets a Boolean value that indicates whether this **Font** object is derived from a GDI vertical font.
Height	Gets the line spacing of this font.
Italic	Gets a value that indicates whether this **Font** object is italic.

Font Properties

Name	Description
Name	Gets the face name of this **Font** object.
Size	Gets the em-size of this **Font** object measured in the unit of this **Font** object.
SizeInPoints	Gets the em-size, in points, of this **Font** object.
Strikeout	Gets a value that indicates whether this **Font** object specifies a horizontal line through the font.
Style	Gets style information for this **Font** object.
Underline	Gets a value that indicates whether this **Font** object is underlined.
Unit	Gets the unit of measure for this **Font** object.

Font Constructor

Public Constructors

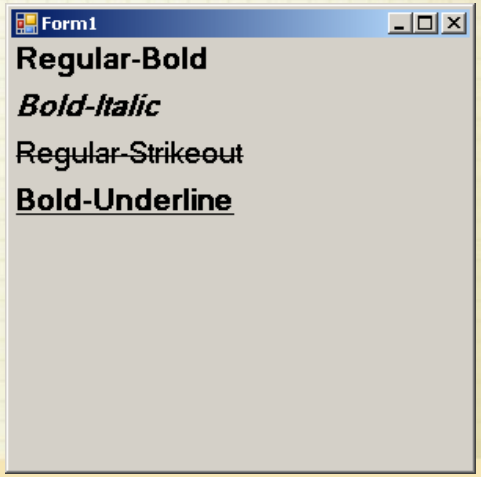
Name	Description
Font	Overloaded. Initializes a new Font that uses the specified existing **Font** and FontStyle.

Font Constructor

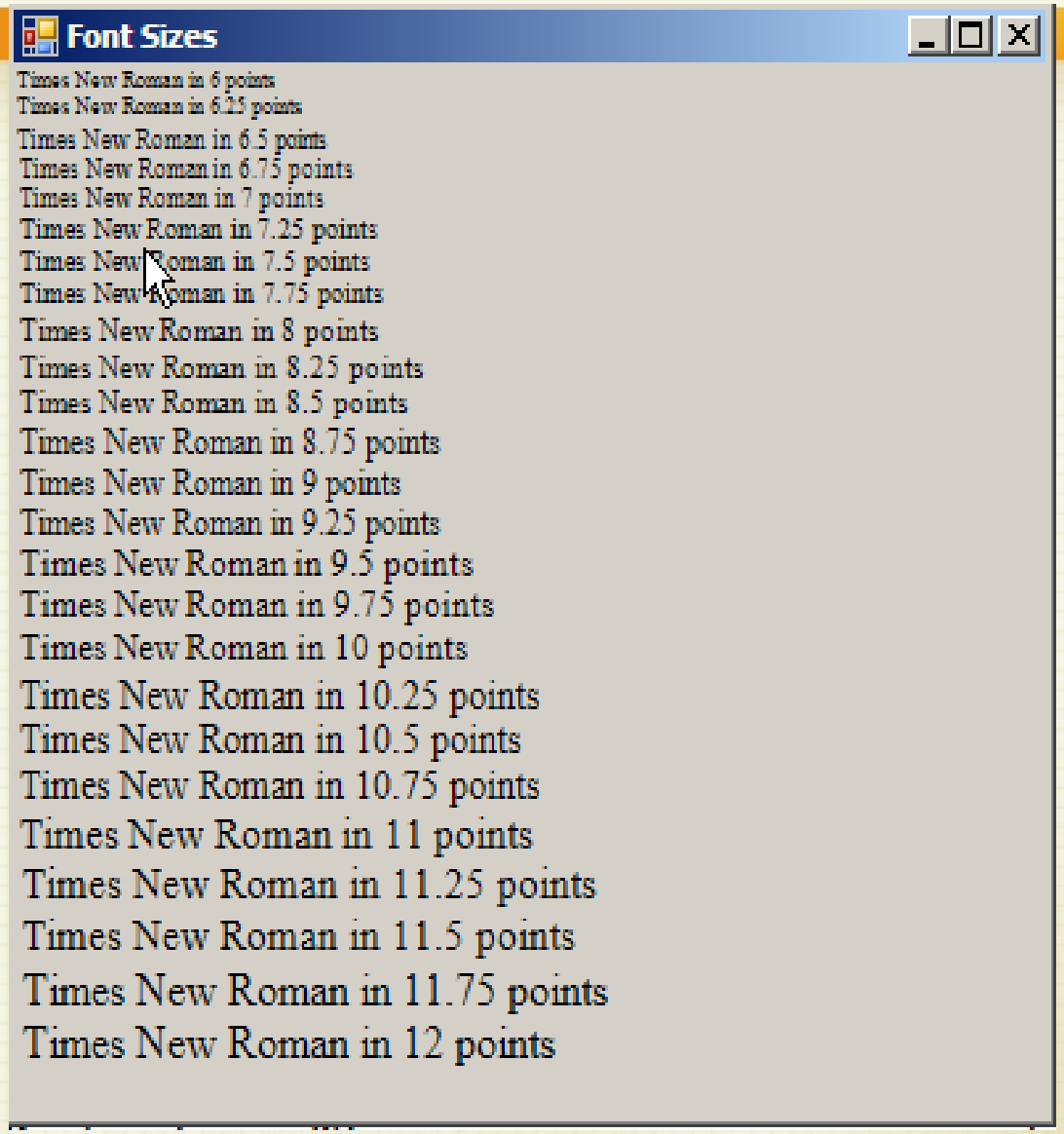
Font(string, float)
Font(string, float, FontStyle)
Font(FontFamily, float)
Font(FontFamily, float, FontStyle)
Font(string, float, GraphicsUnit)
Font(string, float, FontStyle, GraphicsUnit)
Font(FontFamily, float, GraphicsUnit)
Font(string, float, FontStyle, GraphicsUnit, byte, bool)
Font(string, float, FontStyle, GraphicsUnit, byte)
Font(Font, FontStyle)
Font(FontFamily, float, FontStyle, GraphicsUnit)
….

Tập hợp FontStyle

Member Name	Value	Description
Bold	0	Bold text
Italic	1	Italic text
Regular	2	Normal text
Strikeout	4	Text with a line through the middle
Underline	8	Underlined text



Font Sizes


Font Sizes
Times New Roman in 6 points
Times New Roman in 6.25 points
Times New Roman in 6.5 points
Times New Roman in 6.75 points
Times New Roman in 7 points
Times New Roman in 7.25 points
Times New Roman in 7.5 points
Times New Roman in 7.75 points
Times New Roman in 8 points
Times New Roman in 8.25 points
Times New Roman in 8.5 points
Times New Roman in 8.75 points
Times New Roman in 9 points
Times New Roman in 9.25 points
Times New Roman in 9.5 points
Times New Roman in 9.75 points
Times New Roman in 10 points
Times New Roman in 10.25 points
Times New Roman in 10.5 points
Times New Roman in 10.75 points
Times New Roman in 11 points
Times New Roman in 11.25 points
Times New Roman in 11.5 points
Times New Roman in 11.75 points
Times New Roman in 12 points

Font Names

Font Names
Courier New
Courier New
Courier New
Courier New
Arial
Arial
Arial
Arial
Times New Roman
Times New Roman
Times New Roman
Times New Roman

Tập hợp GraphicsUnit

Member Name	Value	Description
World	0	Units of world coordinates
Display	1	Won’t work with font constructor
Pixel	2	Units of pixels
Point	3	Units of 1/72 inch
Inch	4	Units of inch
Document	5	Units of 1/300 inch
Milimeter	6	Units of milimeters

Ví dụ

```
new Font(strFamily, float fSize)
new Font(strFamily, float fSize, GraphicsUnit.Point)
```

```
new Font(strFamily, 72)
new Font(strFamily, 72,    GraphicsUnit.Point)
new Font(strFamily, 1,     GraphicsUnit.Inch)
new Font(strFamily, 25.4f, GraphicsUnit.Millimeter)
new Font(strFamily, 300,   GraphicsUnit.Document)
```

Font Methods

Public Methods

Method	Description
Clone	Creates an exact copy of this Font object.
CreateObjRef	Creates an object that contains all the relevant information required to generate a proxy used to communicate with a remote object.
Dispose	Releases all resources used by this Font object.
Equals	Overridden. Indicates whether the specified object is a Font object and is identical to this **Font** object.
FromHdc	Creates a Font object from the specified Windows handle to a device context.

Font Methods

Public Methods

Method	Description
FromHfont	Creates a Font object from the specified Windows handle.
FromLogFont	Overloaded. Creates a Font object from the specified GDI **LOGFONT** structure.
GetHashCode	Overridden. Gets the hash code for this Font object.
GetHeight	Overloaded. Returns the line spacing of this font.
GetLifetimeService	Retrieves the current lifetime service object that controls the lifetime policy for this instance.

Font Methods

Public Methods

Method	Description
GetType	Gets the Font of the current instance.
InitializeLifetimeService	Obtains a lifetime service object to control the lifetime policy for this instance.
ToHfont	Returns a handle to this Font object.
ToLogFont	Overloaded. Creates a GDI **LOGFONT** structure from this Font object.

Font Methods

Public Methods

Method	Description
ToString	Overridden. Returns a human-readable string representation of this Font object.

Protected Methods

Method	Description
Finalize	Overridden.
MemberwiseClone	Creates a shallow copy of the current Object.

FontFamily Class

- Xác nhận một nhóm các font được thiết kế về cách thể hiện và kiểu dáng tương tự nhau
- Không cho phép kế thừa

- **Namespace:** System.Drawing
- **Assembly:** System.Drawing (in dll)

FontFamily Properties

Public Properties

Name	Description
Families	Returns an array that contains all the FontFamily objects associated with the current graphics context.
GenericMonospace	Gets a generic monospace **FontFamily**.
GenericSansSerif	Gets a generic sans serif **FontFamily** object.
GenericSerif	Gets a generic serif **FontFamily**.
Name	Gets the name of this **FontFamily**.

FontFamily Constructor

Public Constructors

Name	Description
FontFamily	Overloaded. Initializes a new FontFamily that uses the specified name.

Overload List

FontFamily (GenericFontFamilies)
FontFamily (String)
FontFamily (String, FontCollection)

Tập hợp GenericFontFamilies

Member Name	Value	Description
Serif	0	Ví dụ như font **Times New Roman**
SansSerif	1	Ví dụ như font **Arial**
Monospace	2	Ví dụ như font **Courier New**

FontFamily Methods

Public Methods

Name	Description
CreateObjRef	Creates an object that contains all the relevant information required to generate a proxy used to communicate with a remote object
Dispose	Releases all resources used by this FontFamily.
Equals	Determines whether the specified object equals this **FontFamily**.
GetCellAscent	Returns the cell ascent, in design units, of the **FontFamily** of the specified style.
GetCellDescent	Returns the cell descent, in design units, of the **FontFamily** of the specified style.

FontFamily Methods

Public Methods

Name	Description
GetEmHeight	Gets the height, in font design units, of the em square for the specified style.
GetFamilies	Returns an array that contains all the **FontFamily** objects available for the specified graphics context.
GetHashCode	Gets a hash code for this **FontFamily**.
GetLifetimeSer vice	Retrieves the current lifetime service object that controls the lifetime policy for this instance.
GetLineSpacing	Returns the line spacing, in design units, of the **FontFamily** of the specified style.

FontFamily Methods

Public Methods

Name	Description
GetName	Returns the name, in the specified language, of this **FontFamily**.
GetType	Gets the Type of the current instance.
InitializeLifetimeService	Obtains a lifetime service object to control the lifetime policy for this instance.
IsStyleAvailable	Indicates whether the specified FontStyle enumeration is available.
ReferenceEquals	Determines whether the specified Object instances are the same instance.

FontFamily Methods

Public Methods

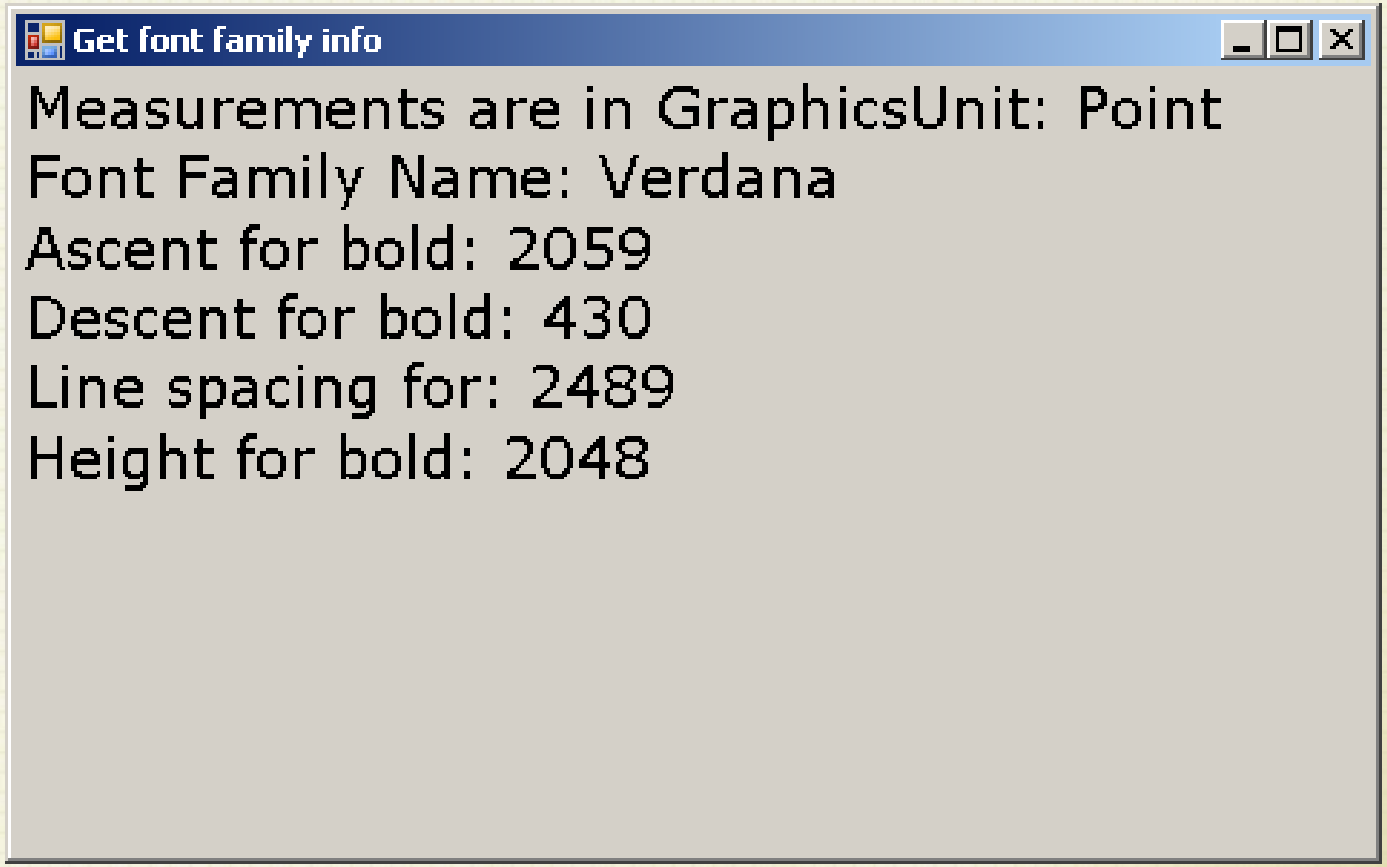
Name	Description
ToString	Converts this **FontFamily** to a human-readable string representation.

Protected Methods

Name	Description
Finalize	Allows an **Object** to attempt to free resources and perform other cleanup operations before the **Object** is reclaimed by garbage collection.
MemberwiseClone	Overloaded.

Ví dụ

Lấy thông tin về font family



Ví dụ

```
using System.Drawing;
using System.Windows.Forms;

private void Form1_Paint(object sender, PaintEventArgs e)
{
    FontFamily myFamily = new FontFamily("Verdana");
    Font myFont = new Font(myFamily, 15, FontStyle.Regular);
    Graphics g = e.Graphics;

    // Print our Family ties...
    string strText = "Measurements are in GraphicsUnit: " +
                                        myFont.Unit.ToString() + "\n" +
                     "Font Family Name: " + myFamily.GetName(0) + "\n" +
                     "Ascent for bold: " +
                         myFamily.GetCellAscent(FontStyle.Bold) + "\n" +
                     "Descent for bold: " +
                        myFamily.GetCellDescent(FontStyle.Bold) + "\n" +
                     "Line spacing for: " +
                        myFamily.GetLineSpacing(FontStyle.Bold) + "\n" +
                     "Height for bold: " +
                                 myFamily.GetEmHeight(FontStyle.Bold);

    g.DrawString(strText, myFont, Brushes.Black, Point.Empty);
    myFont.Dispose();
}
```

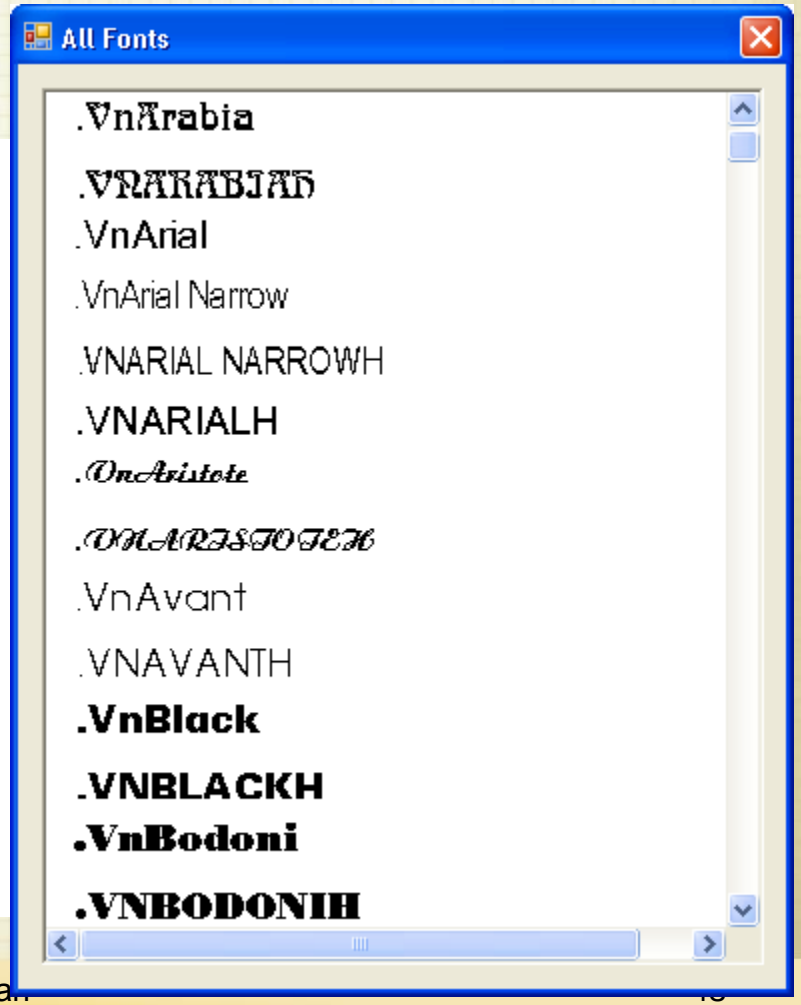
Ví dụ

```
protected void DemoFont(Graphics g)
{
    Fo
    g.                                  10);

    Fo                                  ;
    g.                                  0);

    Fo                                  .Italic);
    g.                                  80);

    Fo                                  ch);
    g.                                  , 120);
}
```

Ví dụ

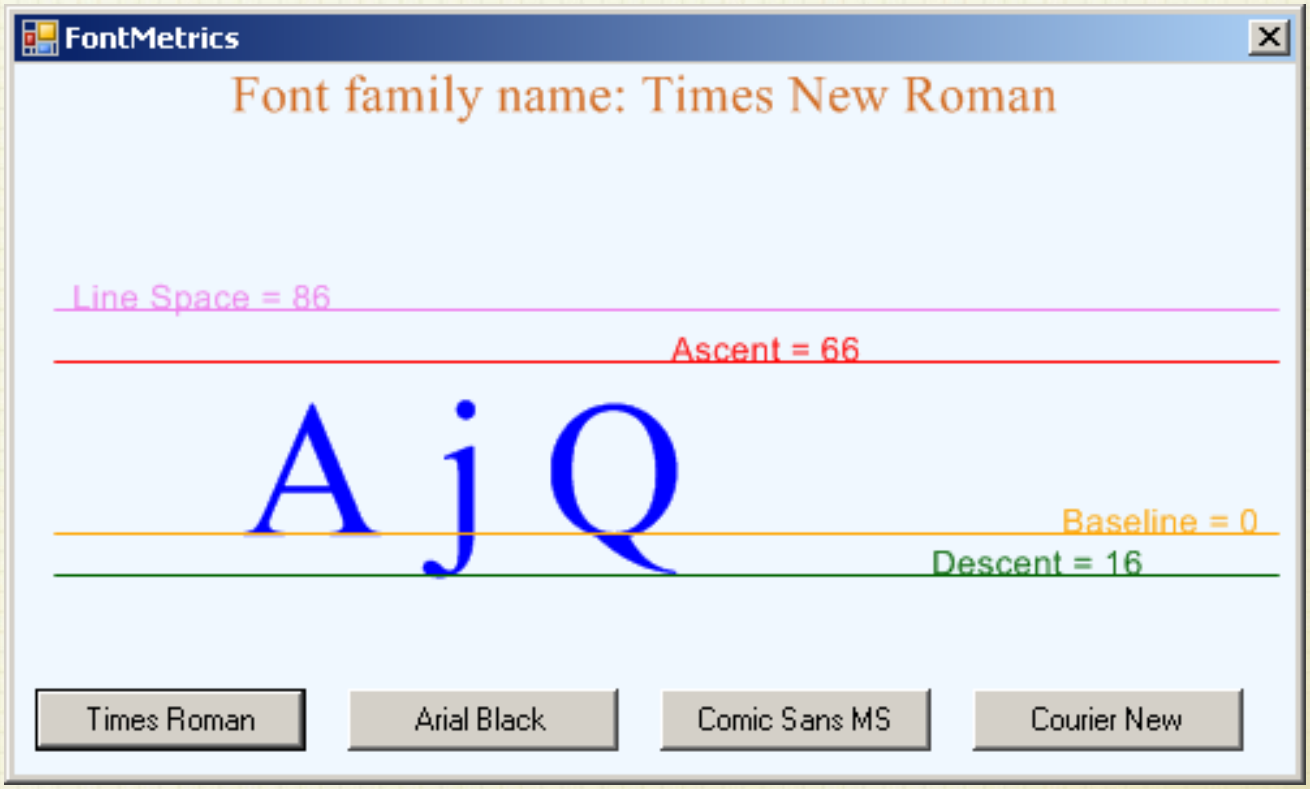
- Lấy danh sách tất cả các font đã được cài đặt trên máy hiện hành

```
InstalledFontCollection fontFamilies =
                    new InstalledFontCollection();
int offset = 10;
foreach (FontFamily family in fontFamilies.Families)
{
    try
    {
        Label fontLabel = new Label();
        fontLabel.Text = family.Name;
        fontLabel.Font = new Font(family, 14);
        fontLabel.Left = 10;
        fontLabel.Width = pnlFonts.Width;
        fontLabel.Top = offset;
        pnlFonts.Controls.Add(fontLabel);
        offset += 30;
    }
    catch { }
}
```

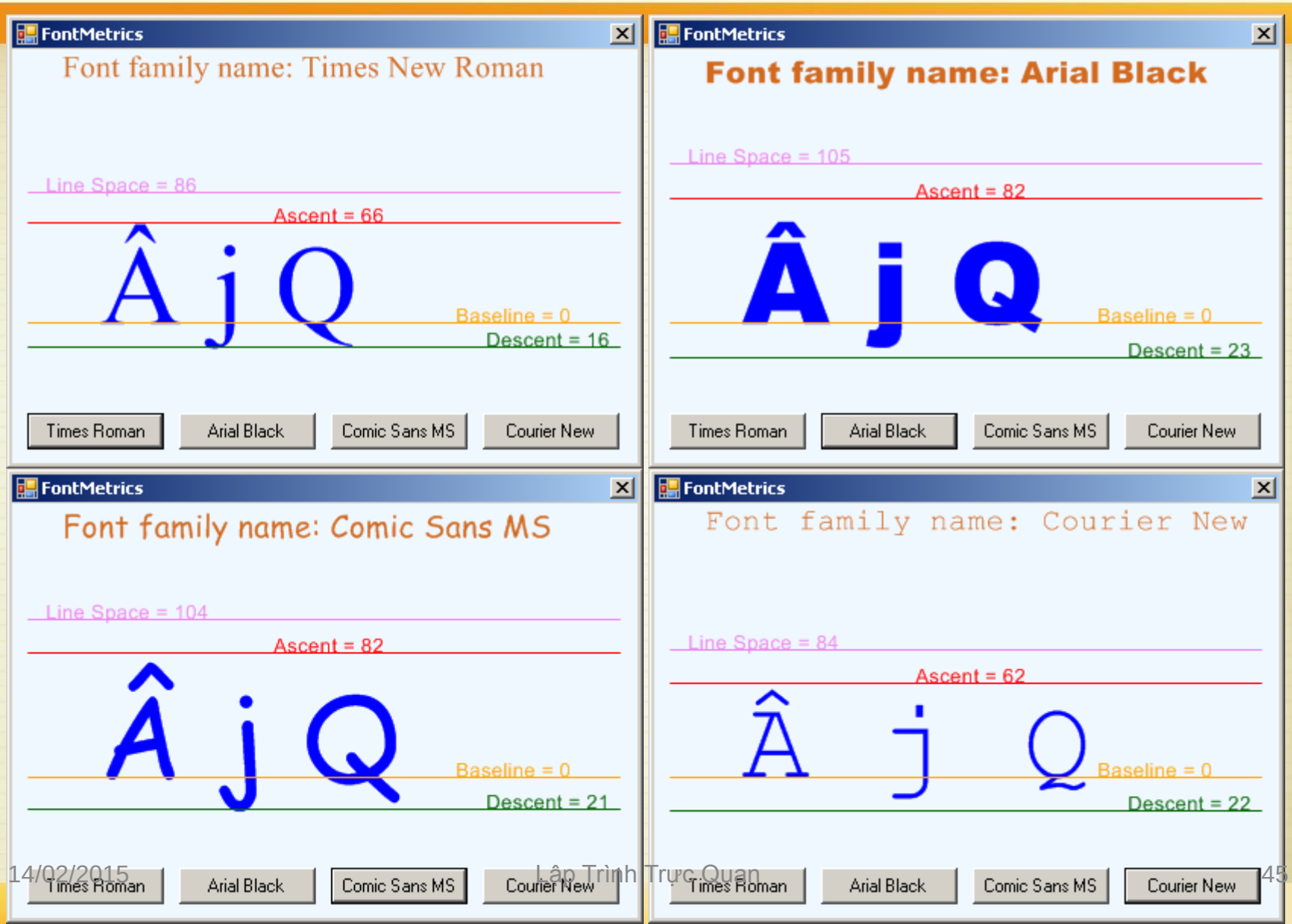


Ví dụ các thông số Họ Font

Vẽ các đường thẳng tại các nơi ascent, descent, baseline, line space



Vẽ các đường thẳng



Vẽ các đường thẳng

```
using System.Drawing;
using System.Drawing.Drawing2D;
using System.Drawing.Text;
using System.Windows.Forms;

namespace FontMetrics_c
{
    public partial class FontMetrics : Form
    {
        private System.Windows.Forms.Button cmdRoman;
        private System.Windows.Forms.Button cmdArial;
        private System.Windows.Forms.Button cmdComic;
        private System.Windows.Forms.Button cmdCourier;

        private void DisplayFontMetrics(FontFamily fontfamily, Font font)
        {
            //Create graphics object and make it pretty
            Graphics G = this.CreateGraphics();
            G.SmoothingMode = SmoothingMode.AntiAlias;
            G.TextRenderingHint = TextRenderingHint.AntiAlias;

            //Get the Design Unit metrics from the font family
            int emSizeInDU = fontfamily.GetEmHeight(font.Style);
            int LineSpaceInDU = fontfamily.GetLineSpacing(font.Style);
            int DescentInDU = fontfamily.GetCellDescent(font.Style);
            int AscentInDU = fontfamily.GetCellAscent(font.Style);
```

Vẽ các đường thẳng

```
//Caculate the Graphics Unit from the font family
float emSizeInGU = font.Size;
int LineSpace = (int)(LineSpaceInDU * emSizeInGU / emSizeInDU);
int Descent = (int)(DescentInDU * emSizeInGU / emSizeInDU);
int Ascent = (int)(AscentInDU * emSizeInGU / emSizeInDU);

//Create the base line to sit the text on
Point BaseLineStart = new Point(10, this.Height * 3 / 5);
Point BaseLineEnd = new Point(this.Width - 20, this.Height * 3 / 5);
//Top left corner of text is the ascent
Point StringPoint = new Point(45, (int)(BaseLineStart.Y - Ascent));

//Clear the screen and draw the string on a base line
G.Clear(Color.AliceBlue);
G.DrawString("Â j Q", font, Brushes.Blue, StringPoint);
G.DrawLine(Pens.Orange, BaseLineStart, BaseLineEnd);

//Draw the annotation lines
Size LineSize = new Size(0, LineSpace);
Size AscentSize = new Size(0, Ascent);
Size DescentSize = new Size(0, Descent);
G.DrawLine(Pens.Violet, BaseLineStart - LineSize, BaseLineEnd - LineSize);
G.DrawLine(Pens.Red, BaseLineStart - AscentSize, BaseLineEnd - AscentSize);
G.DrawLine(Pens.DarkGreen, BaseLineStart + DescentSize, BaseLineEnd + DescentSize);
```

Vẽ các đường thẳng

```
    //Annotate
    Font myfont = new Font(fontfamily, 15);
    G.DrawString("Font family name: " + fontfamily.GetName(0), myfont,
                 Brushes.Chocolate, 30, 0);

    Font AnnoFont = new Font("Arial", 10);
    G.DrawString("Line Space = " + LineSpace.ToString(), AnnoFont,
                Brushes.Violet, 20, (int)(BaseLineStart.Y - LineSpace - 12));

    G.DrawString("Ascent = " + Ascent.ToString(), AnnoFont,
                Brushes.Red, 170, (int)(BaseLineStart.Y - Ascent - 12));

    G.DrawString("Baseline = 0" , AnnoFont,
                Brushes.Orange, 290, (int)(BaseLineStart.Y - 12));

    G.DrawString("Descent = " + Descent.ToString(), AnnoFont,
               Brushes.DarkGreen, 310, (int)(BaseLineStart.Y + Descent -12));
}

private void cmdRoman_Click(object sender, System.EventArgs e)
{
    FontFamily fontfamily = new FontFamily("Times New Roman");
    Font font = new Font(fontfamily, 75, FontStyle.Regular, GraphicsUnit.Pixel);

    DisplayFontMetrics(fontfamily, font);
}
```

Vẽ các đường thẳng

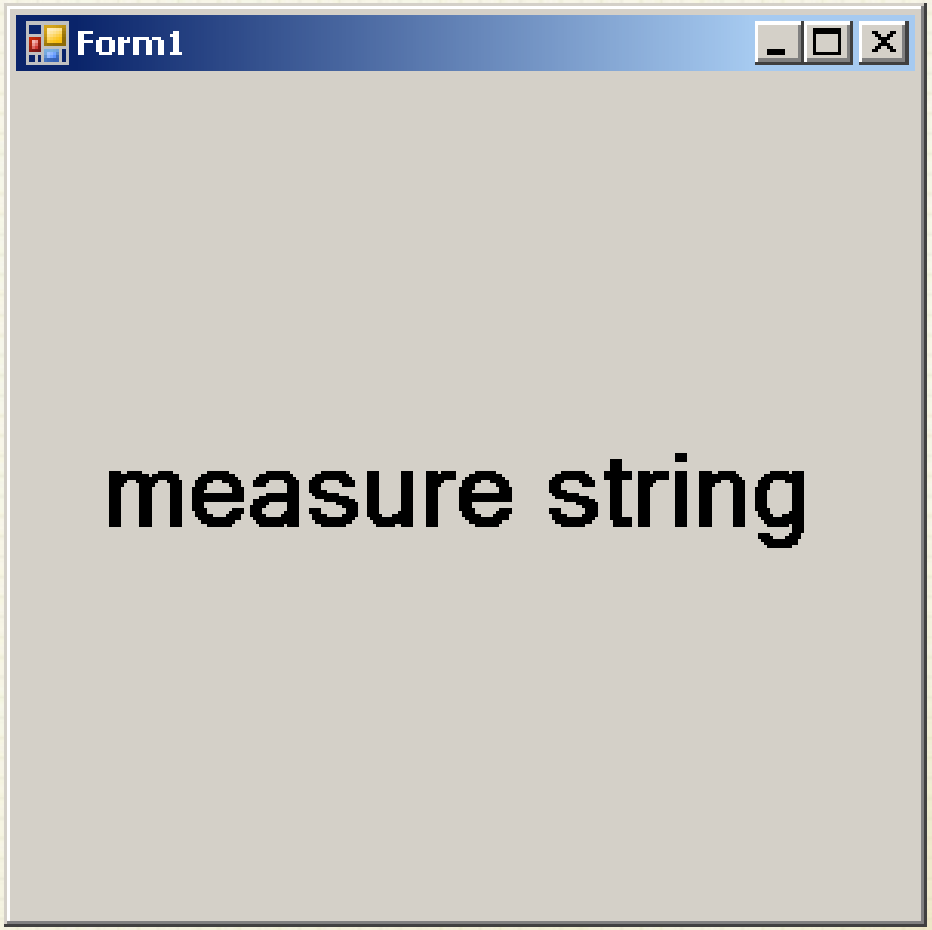
```
    private void cmdArial_Click(object sender, System.EventArgs e)
    {
        FontFamily fontfamily = new FontFamily("Arial Black");
        Font font = new Font(fontfamily, 75, FontStyle.Regular, GraphicsUnit.Pixel);

        DisplayFontMetrics(fontfamily, font);
    }

    private void cmdComic_Click(object sender, System.EventArgs e)
    {
        FontFamily fontfamily = new FontFamily("Comic Sans MS");
        Font font = new Font(fontfamily, 75, FontStyle.Regular, GraphicsUnit.Pixel);

        DisplayFontMetrics(fontfamily, font);
    }

    private void cmdCourier_Click(object sender, System.EventArgs e)
    {
        FontFamily fontfamily = new FontFamily("Courier New");
        Font font = new Font(fontfamily, 75, FontStyle.Regular, GraphicsUnit.Pixel);

        DisplayFontMetrics(fontfamily, font);
    }
  }
}
```

Chuỗi ký tự (String)

- Tập hợp TextRenderingHint
- Tập hợp TextFormatFlags
- Thuộc tính Graphics.TextContrast
- Phương thức Graphics.MeasureString
- Lớp StringFormat
- Lớp TextRenderer

Chuỗi ký tự (String)

Hiện thị chuỗi chính giữa form?



Chuỗi ký tự (String)

- Đo chuỗi khi được vẽ với một loại font cụ thể.

  MeasureString Method (String, Font)

  MeasureString Method (String, Font, int)

- Đo chuỗi khi được vẽ với một loại font cụ thể trong một vùng thể hiện.

  MeasureString Method (String, Font, SizeF)

Chuỗi ký tự (String)

- Đo chuỗi khi được vẽ với 1 loại font cụ thể và được định dạng bởi 1 đối tượng StringFormat.

MeasureString Method (String, Font, int,StringFormat)

MeasureString Method (String, Font, PointF, StringFormat)

MeasureString Method (String, Font, SizeF, StringFormat)

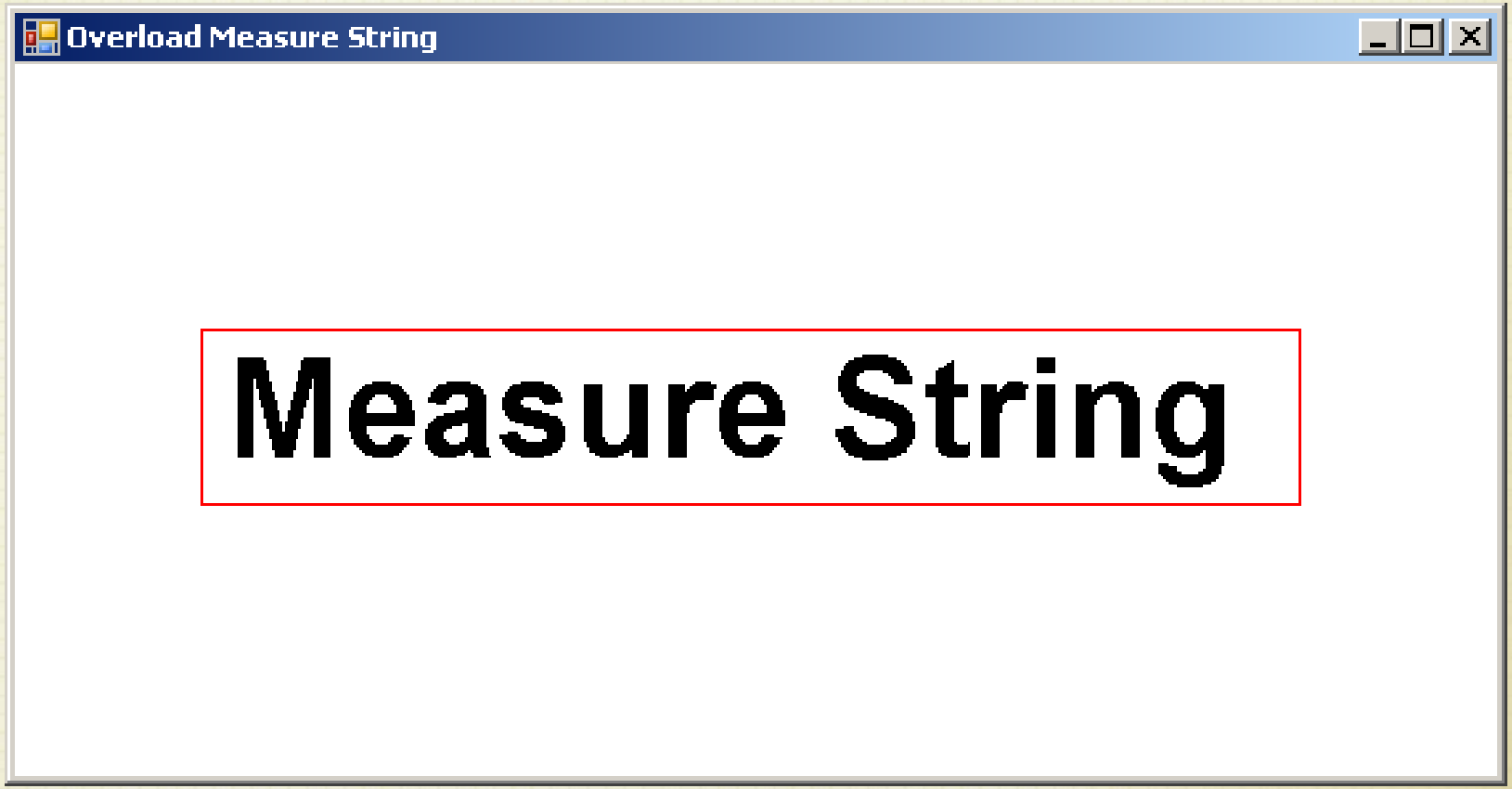
MeasureString Method (String, Font, SizeF, StringFormat, int, int)

Hiển thị chuỗi giữa form

```
using System;
using System.Drawing;
using System.Windows.Forms;

namespace MeasureText
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
        }

        private void Form1_Paint(object sender, PaintEventArgs e)
        {
            Graphics g = e.Graphics;
            Font font = new Font("Time New Roman", 25, FontStyle.Bold);
            string strText = "measure string";
            SizeF sizef = g.MeasureString(strText, font);
            g.DrawString(strText, font, Brushes.Black,
                                (ClientSize.Width - sizef.Width) / 2,
                            (ClientSize.Height - sizef.Height) / 2);
        }
    }
}
```

Ví dụ về hàm MeasureString

Vẽ hình chữ nhật bọc chuỗi



Vẽ hình chữ nhật bọc chuỗi

```
private void Form1_Paint(object sender, PaintEventArgs e)
{
    // Set up string.
    string measureString = "Measure String";
    Font stringFont = new Font("Arial", 40, FontStyle.Bold);
    // Set maximum layout size.
    SizeF layoutSize = new SizeF(500.0F, 500.0F);
    SizeF stringSize = new SizeF();
    //Clear the screen
    e.Graphics.Clear(Color.White);
    //Measure String
    stringSize = e.Graphics.MeasureString(measureString,
                                          stringFont, layoutSize);
    // Draw rectangle representing size of string.
    e.Graphics.DrawRectangle(new Pen(Color.Red, 1), 70.0F, 100.0F,
                             stringSize.Width, stringSize.Height);
    // Draw string to screen.
    e.Graphics.DrawString(measureString, stringFont, Brushes.Black,
                          new PointF(70, 100));
}
```

StringFormat Class

- Chứa tất cả các thông tin về cách thể hiện (layout) của văn bản (như canh dòng, hướng thể hiện, tab stop)
- Không cho phép kế thừa
- **Namespace:** System.Drawing
- **Assembly:** System.Drawing (in dll)

StringFormat Properties

Public Properties

Name	Description
Alignment	Gets or sets text alignment information on the vertical plane.
DigitSubstitutionLanguage	Gets or sets the language that is used when local digits are substituted for western digits.
DigitSubstitutionMethod	Gets or sets the method to be used for digit substitution.
FormatFlags	Gets or sets a StringFormatFlags enumeration that contains formatting information.

StringFormat Properties

Public Properties

Name	Description
GenericDefault	Gets a generic default StringFormat object.
GenericTypographic	Gets a generic typographic **StringFormat** object.
HotkeyPrefix	Gets or sets the HotkeyPrefix object for this **StringFormat** object.
LineAlignment	Gets or sets the line alignment on the horizontal plane.
Trimming	Gets or sets the StringTrimming enumeration for this **StringFormat** object.

Tập hợp StringFormatFlags

Member Name	Description
DirectionRightToLeft	Text is displayed from right to left.
DirectionVertical	Text is vertically aligned.
DisplayFormatControl	Control characters such as the left-to-right mark are shown in the output with a representative glyph.
FitBlackBox	Parts of characters are allowed to overhang the string's layout rectangle. By default, characters are repositioned to avoid any overhang.

Tập hợp StringFormatFlags

Member Name	Description
LineLimit	Only entire lines are laid out in the formatting rectangle. By default layout continues until the end of the text, or until no more lines are visible as a result of clipping, whichever comes first.
MeasureTrailingSpaces	Includes the trailing space at the end of each line. By default the boundary rectangle returned by the MeasureString method excludes the space at the end of each line. Set this flag to include that space in measurement.

Tập hợp StringFormatFlags

Member Name	Description
NoClip	Overhanging parts of glyphs, and unwrapped text reaching outside the formatting rectangle are allowed to show. By default all text and glyph parts reaching outside the formatting rectangle are clipped.
NoFontFallback	Fallback to alternate fonts for characters not supported in the requested font is disabled.
NoWrap	Text wrapping between lines when formatting within a rectangle is disabled. This flag is implied when a point is passed instead of a rectangle, or when the specified rectangle has a zero line length.

Tập hợp StringTrimming

Member Name	Description
Character	Specifies that the text is trimmed to the nearest character.
EllipsisCharacter	Specifies that the text is trimmed to the nearest character, and an ellipsis is inserted at the end of a trimmed line.
EllipsisPath	The center is removed from trimmed lines and replaced by an ellipsis. The algorithm keeps as much of the last slash-delimited segment of the line as possible.
EllipsisWord	Specifies that text is trimmed to the nearest word, and an ellipsis is inserted at the end of a trimmed line.

Tập hợp StringTrimming

Member Name	Description
None	Specifies no trimming.
Word	Specifies that text is trimmed to the nearest word.

Tập hợp HotkeyPrefix

Member Name	Description
Hide	Do not display the hot-key prefix.
None	No hot-key prefix.
Show	Display the hot-key prefix.

StringFormat Constructor

Public Constructors

Name	Description
StringFormat	Overloaded. Initializes a new StringFormat object.

Overload List

StringFormat ()
StringFormat (StringFormat)
StringFormat (StringFormatFlags)
StringFormat (StringFormatFlags, Int32)

StringFormat Methods

Public Methods

Name	Description
Clone	Creates an exact copy of this StringFormat object.
CreateObjRef	Creates an object that contains all the relevant information required to generate a proxy used to communicate with a remote object.
Dispose	Releases all resources used by this **StringFormat** object.
Equals	Determines whether two Object instances are equal.
GetHashCode	Serves as a hash function for a particular type.

StringFormat Methods

Public Methods

Name	Description
GetLifetimeService	Retrieves the current lifetime service object that controls the lifetime policy for this instance.
GetTabStops	Gets the tab stops for this **StringFormat** object.
GetType	Gets the Type of the current instance.
InitializeLifetimeService	Obtains a lifetime service object to control the lifetime policy for this instance.

StringFormat Methods

Public Methods

Name	Description
ReferenceEquals	Determines whether the specified **Object** instances are the same instance.
SetDigitSubstitution	Specifies the language and method to be used when local digits are substituted for western digits.
SetMeasurableCharacterRanges	Specifies an array of CharacterRange structures that represent the ranges of characters measured by a call to the MeasureCharacterRanges method.
SetTabStops	Sets tab stops for this **StringFormat** object.

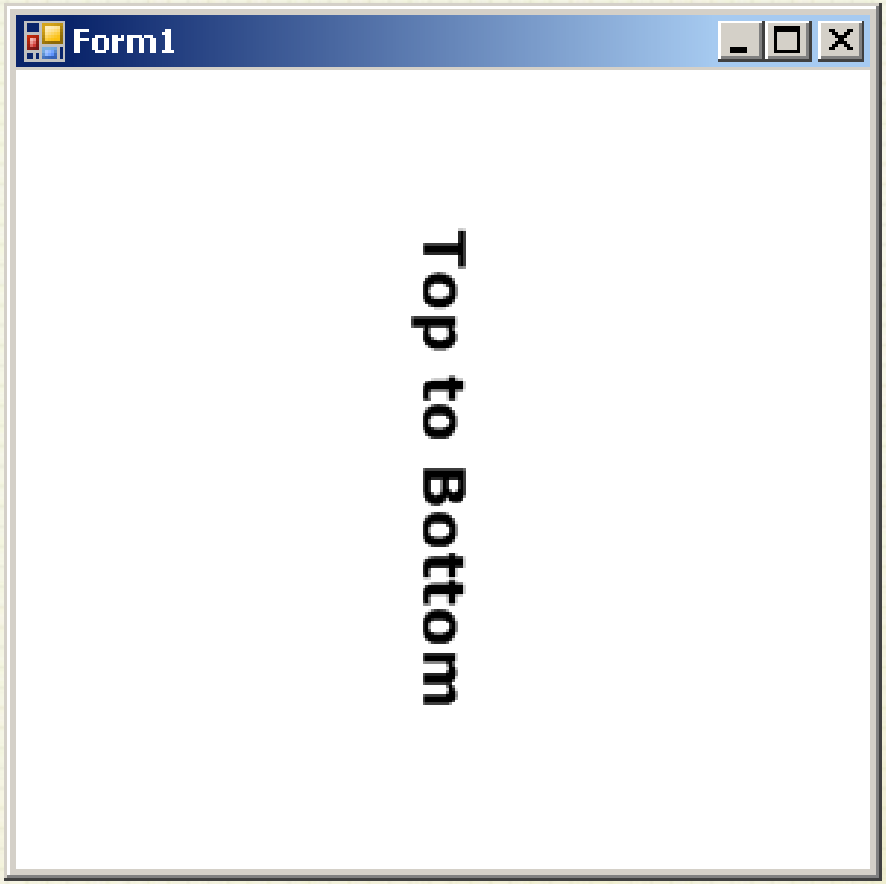
StringFormat Methods

Protected Methods

Name	Description
Finalize	Allows an **Object** to attempt to free resources and perform other cleanup operations before the **Object** is reclaimed by garbage collection.
MemberwiseClone	Overloaded.

Ví dụ về StringFormat

Hiển thị chuỗi theo chiều thẳng đứng

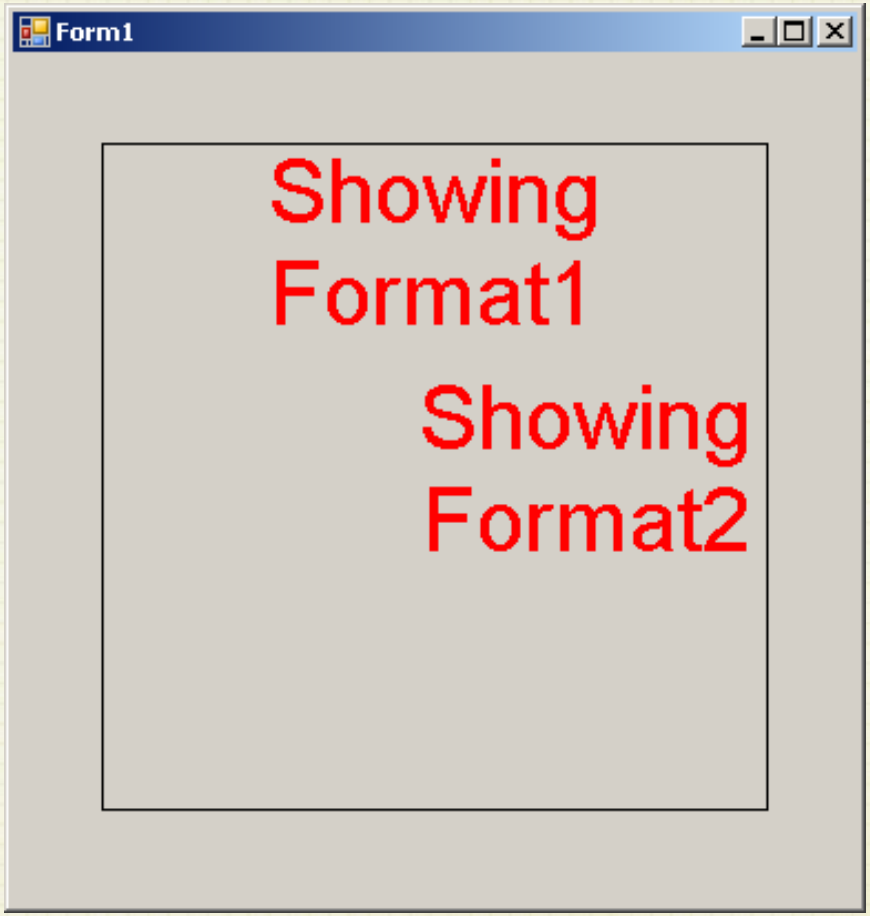


Ví dụ về StringFormat

```
using System.Drawing;
using System.Drawing.Text;
using System.Windows.Forms;

private void Form1_Paint(object sender, PaintEventArgs e)
{
    string strText = "Top to Bottom";
    Graphics g = e.Graphics;
    Font font = new Font("Verdana", 15, FontStyle.Bold);

    // Set string format.
    StringFormat strFormat = new StringFormat();
    strFormat.FormatFlags = StringFormatFlags.DirectionVertical;

    strFormat.Alignment = StringAlignment.Center;
    strFormat.LineAlignment = StringAlignment.Center;

    // Clear the screen
    g.Clear(Color.White);
    g.TextRenderingHint = TextRenderingHint.AntiAlias;

    // Draw string to screen.
    g.DrawString(strText, font, Brushes.Black,
      new RectangleF(0, 0, ClientRectangle.Width, ClientRectangle.Height), strFormat);
}
```

Ví dụ về StringFormat

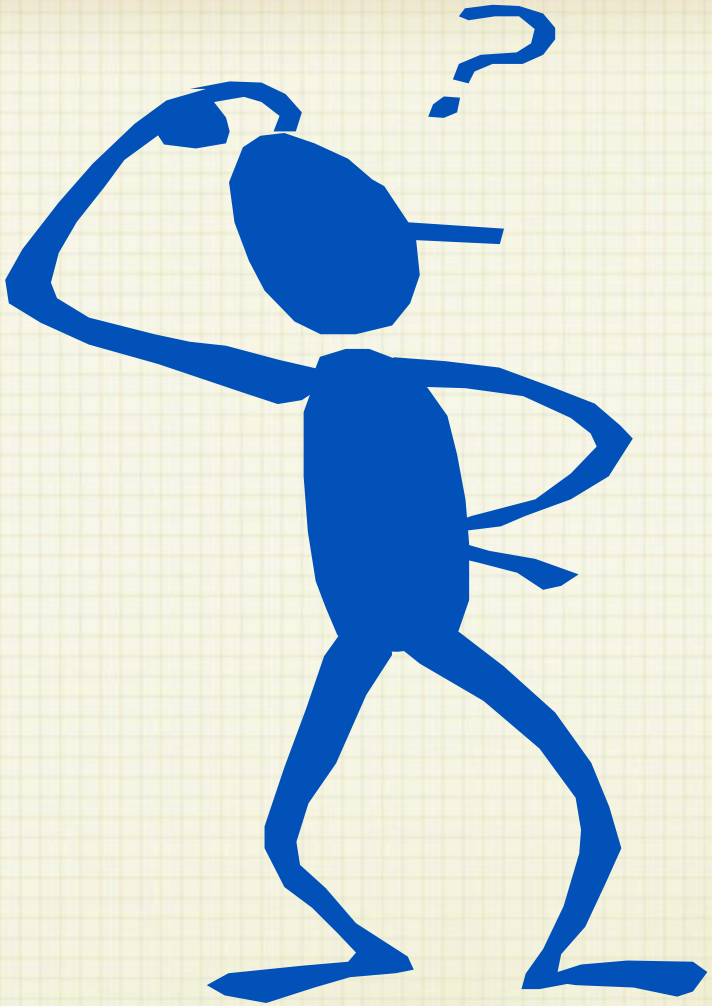
Hiển thị chuỗi và canh lề



Ví dụ về StringFormat

```
using System.Drawing;
using System.Windows.Forms;

private void Form1_Paint(object sender, PaintEventArgs e)
{
    Font MyFont = new Font("Arial",30);
    // Construct a new Rectangle .
    Rectangle displayRectangle =
       new Rectangle(new Point(40, 40), new Size(300, 300));

    // Construct 2 new StringFormat objects
    StringFormat format1 = new StringFormat(
                                StringFormatFlags.NoClip);
    StringFormat format2 = new StringFormat(format1);

    // Set the LineAlignment and Alignment properties for
    // both StringFormat objects to different values.
    format1.LineAlignment = StringAlignment.Near;
    format1.Alignment = StringAlignment.Center;
    format2.LineAlignment = StringAlignment.Center;
    format2.Alignment = StringAlignment.Far;
```

Ví dụ về StringFormat

```
    // Draw the bounding rectangle and a string for each
    // StringFormat object.
    e.Graphics.DrawRectangle(Pens.Black, displayRectangle);
    e.Graphics.DrawString("Showing Format1", MyFont,
        Brushes.Red, (RectangleF)displayRectangle, format1);
    e.Graphics.DrawString("Showing Format2", MyFont,
        Brushes.Red, (RectangleF)displayRectangle, format2);

    MyFont.Dispose();
}
```

Q & A